

CE 490 – Introduction to Digital Image Processing

TERM-PROJECT

FALL 2025-2026

1. Overview

This project is an essential component of CE 490 – Introduction to Digital Image Processing. Students will design and implement their own image processing algorithm using MATLAB that demonstrates their expertise on key topics covered in the course.

2. Objectives

Students will apply theoretical concepts to solve image processing problems and get hands-on experience in MATLAB by implementing image processing algorithms. They will analyze, assess, and present image processing results efficiently.

3. Tools

MATLAB (Image Processing Toolbox recommended)

4. Suggested Input Data

All input images are provided in a zip file along with this project description document. There are five image pairs containing **the original (clean) and degraded (noisy and blur)** versions:

- **[512x512 pixels]**- original_boat.png & degraded_boat.png
- **[512x512 pixels]**- original_baboon.png & degraded_baboon.png
- **[512x512 pixels]**- original_barbara.png & degraded_barbara.png
- **[256x256 pixels]**- original_cameraman.png & degraded_cameraman.png
- **[384x512 pixels]**- original_peppers.png & degraded_peppers.png

5. Project Description

The project simulates real-world scenarios where **images are corrupted by noise and blur**. Students will implement **their own** MATLAB-based algorithm to **restore and enhance the degraded images given**. The project content must include **a minimum of three main methods learned during the semester**.

6. Evaluation and Analysis

Students will compare original, degraded, and restored images using **quantitative metrics**:

- **PSNR** (Peak Signal-to-Noise Ratio),
- **MSE** (Mean Squared Error),
- **SSIM** (Structural Similarity Index).

Students will construct statistical comparison tables for all images and discuss the results in detail. They will also inspect images visually and make side-by-side comparisons.

7. Project Timeline

- **Project Proposal Submission (Week 10):** 1-2 pages describing the title, objective, literature background, methods planned, expected outcomes, and evaluation plan.
- **Presentation (Weeks 13-14-15):** 10-15 minutes per group with demonstration of MATLAB results and visualization.
- **Final Report and Code Submission (Week 16):** A full report describing the **Abstract, Introduction, Methodology, Results, and Conclusion**. MATLAB code should be with well-commented scripts and/or functions implementing the complete processing pipeline.

8. Academic Honesty and Use of AI Tools

All submitted work must be your own. Students are free to talk over ideas with others but must present their own written reports and code. The use of MATLAB built-in functions is allowed and encouraged where appropriate. AI tools (such as ChatGPT or MATLAB Copilot) may be used for guidance, debugging, or to clarify concepts; however, students must not present AI-generated text or code as their own without appropriate modification and comprehension. **Any use of AI tools must be documented in the final report under a 'Tools and Resources Used' heading.**